ANSWERS

Problem 1

a) A rigid column AE has a maximum buckling load. Replace the beam DEC by a rotational spring with stiffness r at E. This model is a basic case for a rigid column with a rotational spring:

$$r = \frac{3EI_2}{3} + \frac{3EI_2}{4} = \frac{21EI_2}{12} = \frac{21 \times 3000}{12} = 5250 \text{ kNm/rad}$$

$$F_{c-rigid} = \frac{r}{l} = \frac{5250}{5} = 1050 \text{ kN}$$

b) For a flexible column AE, a sway-model is used with a rotational spring at E. The buckling load can be found with the eta-formula:

$$F_c = \frac{1}{\eta}\frac{\pi^2 EI_1}{5^2} \quad \text{with: } \rho = \frac{5250 \times 5}{EI_1} \quad \text{and} \quad \eta = 4 + \tfrac{10}{\rho}$$

$$F_c = \frac{\pi^2 EI_1}{25\left(4 + \dfrac{EI_1}{2625}\right)}$$

This load is 54.6 % of the previous found value of 1050 kN. This results in a bending stiffness EI_1 for column AE:

$$\frac{\pi^2 EI_1}{25\left(4 + \dfrac{EI_1}{2625}\right)} = 0.546 \times 1050 \quad \Rightarrow \quad EI_1 = 13001 \text{ kNm}^2$$

c) The load mF acts as a leaning load (aanpendelend) and increases the effective compressive load on column AE in the deformed situation. This effect can be estimated with:

$$F_{tot}^{\text{AE}} = F + mF\frac{l^{\text{AE}}}{l^{\text{BD}}} = \left(1 + m \times \frac{5}{2.5}\right)F = (1 + 2m)F$$

d) The maximum value of F follows from:

$$1.5F \le 0.546 \times 1050 \quad \Rightarrow \quad F_{\max} = 383.2 \text{ kN}$$

e) The 1st order displacement at E from a linear approach results in:

$$u_1^{\text{E}} = \frac{H \times 5^3}{3 \times 13001} + \frac{(H \times 5)}{5250} \times 5 = 0.04 \text{ m}$$

f) De 2nd order displacement at E follows from:

$$u_2^{\text{E}} = \frac{n}{n-1}u_1^{\text{E}} \quad \text{with: } n = \frac{382.2}{1.5 \times 50} = 5.096$$

$$u_2^{\text{E}} = 1.244\, u_1^{\text{E}} = 0.05 \text{ m} \quad \text{Remark: This enlargement factor is unacceptable!}$$